

Computational Science and Engineering (CSE) Minor

Final Minor Project Proposal (Parts A & B)

Courses: TW3715TU & TW3725TU



Section 1: Administrative Details

- **Project Title:** A Divergence-Free Extension of the (Generalized/Weighted) Shifted Boundary Method for Immersed Moving-Boundary Stokes Flow
 - **Supervisor(s):**
 - Oriol Colomés (j.o.colomesgene@tudelft.nl)
 - Shreyas Prashanth (s.prashanth-1@tudelft.nl)
 - **Faculty & Department:**
 - Civil Engineering and Geosciences, Hydraulic Engineering, Offshore Engineering section (<https://www.tudelft.nl/citg/over-faculteit/afdelingen/hydraulic-engineering/sections/offshore-engineering>)
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Section 2: Core Focus & Alignment

Main Focus of the Project

- Modeling with numerical methods

Minor Course Alignment

(Please keep the courses that are directly related to this project and delete the rest)

- **TW3710TU** Scientific Programming
- **TW3730TU** Numerical Methods for Differential Equations

Required Prerequisite Knowledge

- **Physics:** Fundamentals of fluid dynamics is a plus (notion of incompressible flows)
 - **Engineering:** no background required
 - **Mathematics:** Finite element methods, fundamentals of functional analysis is a plus (otherwise it will be part of literature review phase)
 - **Programming Languages:** Julia (easy to learn for python programmers)
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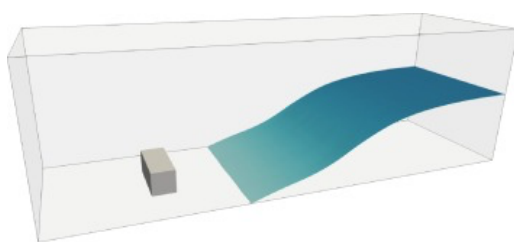


Section 3: Project Description

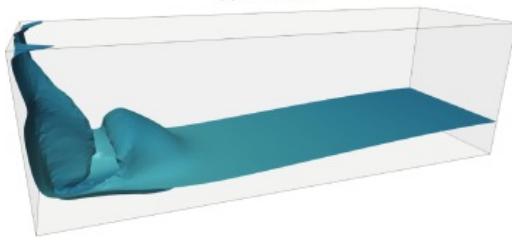
Submission Limit: The entire description (including Parts A and B below) must not exceed **2 A4 pages**. Supervisors are highly encouraged to include relevant images, diagrams, or links to online content.

Overview

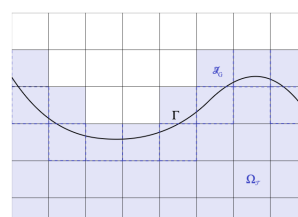
Our group develops unfitted (immersed) finite element methods for PDEs on complex and moving geometries. These are methods that let you solve a given PDE on a fixed background mesh without ever having to re-mesh as a boundary moves or a shape changes. One of the main topics where we are working on is the simulation of fluid flows around moving structures, in the context of fluid-structure interaction in marine environments. However, simulating fluid flow around immersed objects is a recurring challenge in computational fluid dynamics, not only in offshore engineering applications, e.g. a valve opening, an oscillating cylinder, a ship hull, or a growing/shrinking interface.



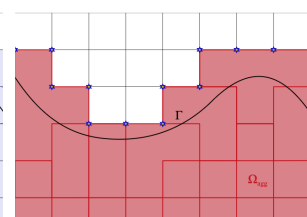
(a) $t = 0.3s$.



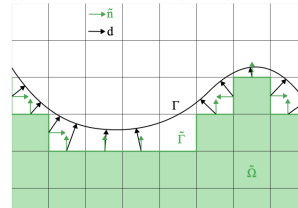
(c) $t = 1.1s$.



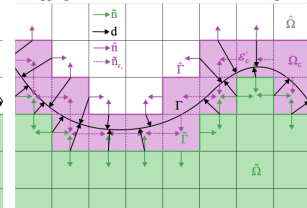
(a) Cut Finite Element Method (CutFEM)



(b) Aggregated unfitted Finite Element Method (AgFEM)



(c) Shifted Boundary Method (SBM)



(d) Weighted Shifted Boundary Method (WSBM)

The class of Shifted Boundary-based Methods (SBM) is a modern unfitted finite element technique that avoids this problem. Instead of cutting elements exactly at the true boundary, SBM works with a nearby surrogate boundary made of whole mesh edges/faces, and corrects for the small geometric mismatch using a Taylor expansion ("shift") of the boundary condition. One of the main issues of this approach is the lack of mass conservation when geometries move across the background grid. We have the hypothesis that using a divergence-free finite element discretization will alleviate this issue. This is precisely the goal of the study: to derive and implement a formulation of the shifted boundary method using divergence-free finite elements. Note that the implementation of such type of elements is already available in the software that will be used in the project, Gridap.jl (<https://github.com/gridap/Gridap.jl>), the work will focus on the derivation of the formulation and posterior implementation.

Part A: Literature Study (TW3715TU | 2 ECTS)

- **Key Introductory Material:**
 - Main, Alex, and Guglielmo Scovazzi. "The shifted boundary method for embedded domain computations. Part I: Poisson and Stokes problems." *Journal of Computational Physics* 372 (2018): 972-995.
 - **Further Required Reading:**
 - Colomés, Oriol, et al. "A weighted shifted boundary method for free surface flow problems." *Journal of Computational Physics* 424 (2021): 109837.
 - Frachon, Thomas, Erik Nilsson, and Sara Zahedi. "Divergence-free cut finite element methods for Stokes flow." *BIT Numerical Mathematics* 64.4 (2024): 39.
 - **Core Concepts & Focus Points:**
 - *Learning unfitted Finite-Element Methods and learn how to implement FE formulations in user-friendly FE package.*
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Part B: Main Project Implementation (TW3725TU | 6 ECTS)

- **Project Stages & Milestones:**
 - Reproduce the baseline (1-2 weeks). Implement the existing (G/W)SBM formulation for the time-dependent Stokes problem with an immersed/moving circular boundary, using standard continuous $\mathbb{Q}_2$ - $\mathbb{Q}_1$ (Taylor–Hood-type) elements, reproducing a subset of the published benchmarks (manufactured-solution convergence; oscillating cylinder/square pressure-oscillation study). Baseline implementation already available for comparison.

- Derive formulation and implement the divergence-free variant (1-2 weeks). Replace the velocity space with a lowest-order Raviart–Thomas space (RT_0 or RT_1) paired with piecewise-constant/piecewise-linear pressure and keep the shifted-boundary Nitsche treatment of the Dirichlet condition.
- Verify exact discrete divergence-freedom numerically (1-2 weeks). Confirm $\|\nabla \cdot \mathbf{u}_h\|$ is at (or near) machine precision in every cell of the background mesh for both static and moving geometries.
- Compare pressure-oscillation behavior against the baseline (1-2 weeks). Repeat (a scaled-down version of) the oscillating-cylinder-in-cross-flow benchmark and quantify whether the RT-based scheme removes the Δt^{-1} growth of pressure oscillations seen in the baseline (G/W)SBM.

- **Expected Outcomes:**

- Working code (Julia/Gridap.jl), version-controlled in a public github repository, documented, and reproducible from a clean environment.
- Numerical verification suite: manufactured-solution convergence plots (static case), discrete divergence-free cell-wise plot, pressure-oscillation comparison against baseline GSBM/WSBM on at least one moving-boundary benchmark.